KON426E Intelligent Control Systems

Assignment 2 Report

Atakan Kuşçu
Student ID: 040210520

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Abstract

This report presents the solution of Assignment 2 for KON426E Intelligent Control Systems. Three intelligent control and system identification problems are studied. First, a nonlinear black-box discrete-time system is identified using a NARX neural network. The delay order is selected through validation on a smooth operational input, and the final model achieves a very low prediction error. Second, a nonlinear spherical tank level-control problem is investigated. A fixed-gain PI controller is first tested at different operating points to demonstrate performance degradation due to the nonlinear tank geometry. Then, a fuzzy gain-scheduled PI controller is implemented to adapt the controller gains online. Finally, a fuzzy supervisory speed controller is designed for waypoint navigation of an unmanned surface vessel. The fuzzy controller successfully prevents the orbiting behavior observed with a constant-speed baseline controller.

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1 Question 1: Neural System Identification

1.1 Problem Description

The first part of the assignment considers a nonlinear black-box discrete-time dynamic plant. The physical governing equations of the plant are unknown. Therefore, a data-driven system identification approach is required. The plant is represented by the compiled MATLAB model `System1.p`. The system is known to be representable by a NARX structure, in which the next output depends on current and past input samples as well as past output samples.

The general NARX representation used in this study is

$$\hat{y}(k+1) = F\left(u(k), u(k-1), \ldots, u(k-d), y(k), y(k-1), \ldots, y(k-d+1)\right), \qquad (1)$$

where d is the delay order and $F(\cdot)$ is approximated by a neural network.

1.2 Data Generation and Excitation

A training input signal with 3000 samples was generated using uniformly distributed random noise in the interval $[-1.5, 1.5]$:

$$u_{\text{train}}(k) \sim \mathcal{U}(-1.5, 1.5), \qquad k = 1, 2, \ldots, 3000. \qquad (2)$$

The black-box model was reset before simulation to ensure an independent run, and the resulting output sequence y_{train} was collected.

Random excitation is suitable for nonlinear system identification because it excites the plant over a broad range of amplitudes and frequencies. In contrast, a simple step input provides information only around a limited operating region and does not sufficiently reveal the dynamic response of the system under varying input conditions. Therefore, the random input signal provides richer information for training the NARX model.

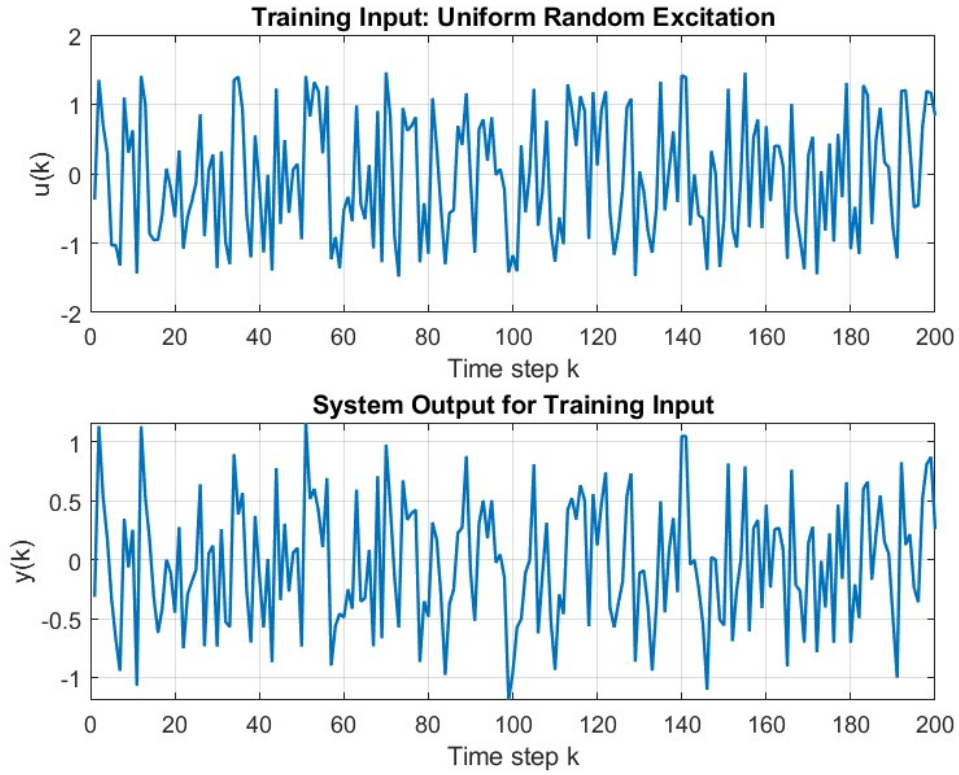


Figure 1: First 200 samples of the random training input and the corresponding system output.

1.3 NARX Network Architecture and Training

A MATLAB `narxnet` architecture was used for system identification. Since the plant update function computes the next output using the current scalar input, the input delay vector was selected as

$$\text{inputDelays} = 0 : d, \tag{3}$$

whereas the feedback delay vector was selected as

$$\text{feedbackDelays} = 1 : d. \tag{4}$$

This is important because using only $1 : d$ for the input delays would exclude the current input $u(k)$, even though the system output $y(k+1)$ directly depends on it.

The candidate delay orders were tested over a range of values, and the final delay was selected according to the validation RMSE on the operational input profile. The selected NARX structure was

- Input delays: $0 : 6$,
- Feedback delays: $1 : 6$,
- Hidden layers: two hidden layers with 20 and 10 neurons,

- Hidden-layer activation functions: hyperbolic tangent sigmoid, `tansig`,
- Output activation function: linear, `purelin`,
- Training algorithm: Levenberg–Marquardt, `trainlm`.

The selected delay order was $d = 6$. The delay selection result is summarized in Table 1.

Table 1: Selected NARX model configuration.

Quantity	Selected value
Best delay order	$d = 6$
Input delays	$0 : 6$
Feedback delays	$1 : 6$
Hidden layers	$[20, 10]$
Training algorithm	Levenberg–Marquardt
Best one-step validation RMSE	0.003173

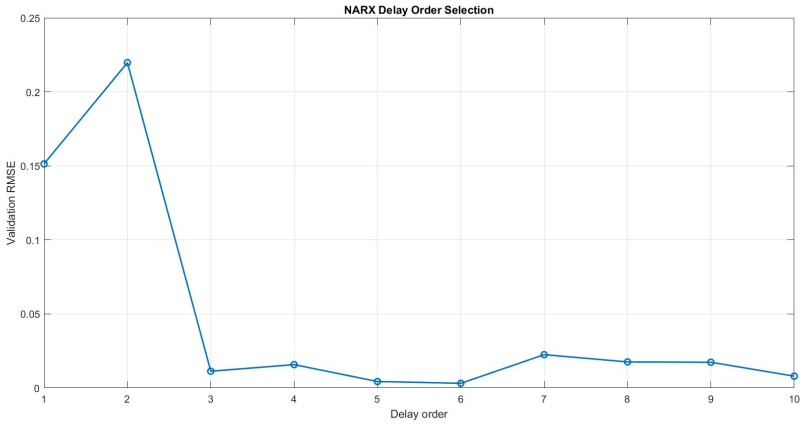


Figure 2: Validation RMSE versus NARX delay order. The best delay order was found as $d = 6$.

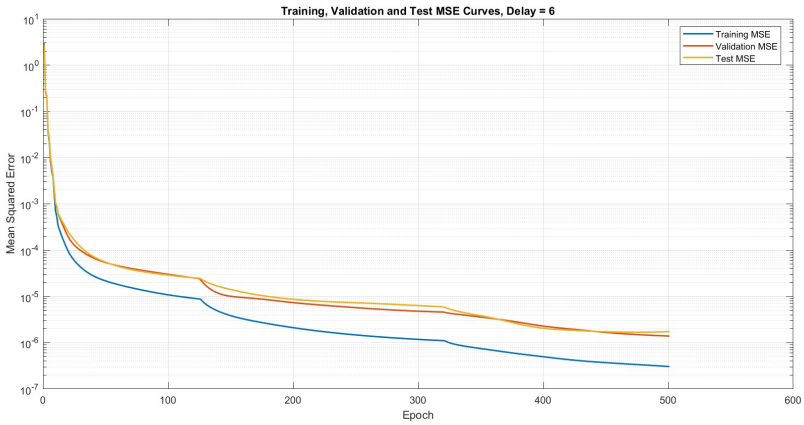


Figure 3: Training, validation, and test MSE curves of the selected NARX network.

1.4 Validation on an Operational Profile

The trained NARX model was validated using the smooth operational input

$$u_{\text{test}}(k) = \sin\left(\frac{2\pi k}{25}\right) + 0.5\sin\left(\frac{2\pi k}{10}\right), \qquad k = 1, 2, \ldots, 500. \tag{5}$$

Before generating the true plant output, the black-box system was reset. The trained network was then simulated on the same input sequence. Both one-step-ahead prediction and closed-loop prediction were evaluated.

The RMSE was computed as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(y(k) - \hat{y}(k)\right)^2}. \tag{6}$$

Table 2: Operational validation results of the selected NARX model.

Prediction mode	RMSE
One-step-ahead prediction	0.003173
Closed-loop prediction	0.002850

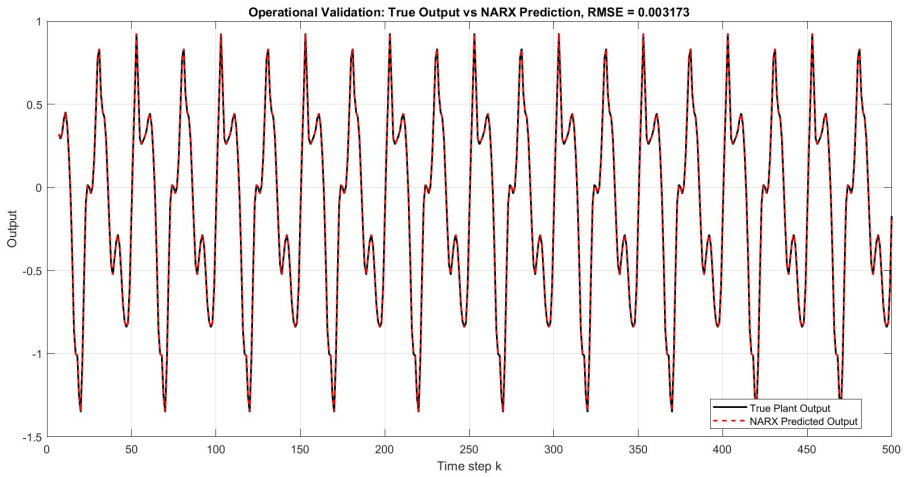


Figure 4: Operational validation: true plant output and NARX-predicted output.

The true output and the predicted output almost completely overlap. The very low RMSE values indicate that the selected NARX network accurately identified the nonlinear dynamic behavior of the black-box system. The result also confirms that the delay order was selected appropriately.

2 Question 2: FGS-PI Control of a Nonlinear Spherical Tank

2.1 Problem Description and Plant Model

The second problem considers fluid-level control of a nonlinear spherical tank. The manipulated variable is the inflow rate $u(t)$, and the controlled variable is the fluid height $h(t)$. The nonlinear plant dynamics are obtained from conservation of mass:

$$A(h(t))\frac{dh(t)}{dt} = u(t) - C_v\sqrt{h(t)}. \tag{7}$$

The height-dependent cross-sectional area of the spherical tank is

$$A(h) = \pi(2Rh - h^2). \tag{8}$$

Thus, the state equation is

$$\dot{h} = \frac{u - C_v\sqrt{h}}{\pi(2Rh - h^2)}. \tag{9}$$

The parameters used in the simulation are listed in Table 3.

Table 3: Spherical tank parameters.

Parameter	Symbol	Value
Tank radius	R	1.5 m
Outlet coefficient	C_v	0.2
Maximum tank height	$h_{\max}$	3.0 m

At $h = 0$, the cross-sectional area becomes $A(0) = 0$, which makes the differential equation singular. Therefore, the simulation was initialized at $h_0 = 0$ as required, but a small numerical regularization

$$h_{\text{eff}} = \max(h, \varepsilon), \qquad \varepsilon = 10^{-6}, \tag{10}$$

was used only inside the area and outflow calculations. This prevents division by zero while preserving the intended empty-tank operating condition.

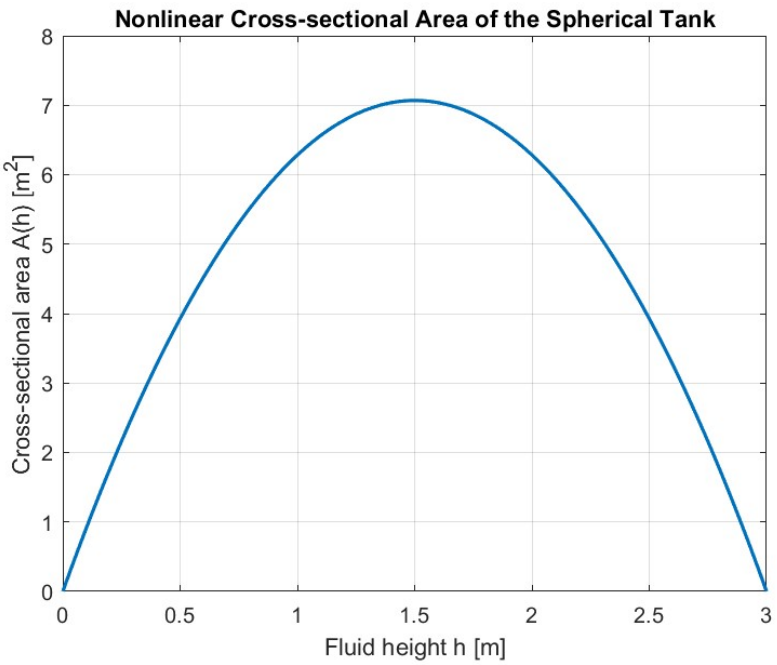


Figure 5: Nonlinear cross-sectional area $A(h)$ of the spherical tank. The area is largest near the middle of the tank.

2.2 Baseline Fixed-Gain PI Control

A standard fixed-gain PI controller was first implemented:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau, \tag{11}$$

where

$$e(t) = h_{\text{ref}}(t) - h(t). \tag{12}$$

The controller gains were tuned for a fast and stable response for the low-level step from $h = 0.0$ m to $h = 0.5$ m. The selected fixed gains were

$$K_p = 15.0, \qquad K_i = 0.30. \tag{13}$$

The same gains were then tested for the step from $h = 1.0$ m to $h = 1.5$ m.

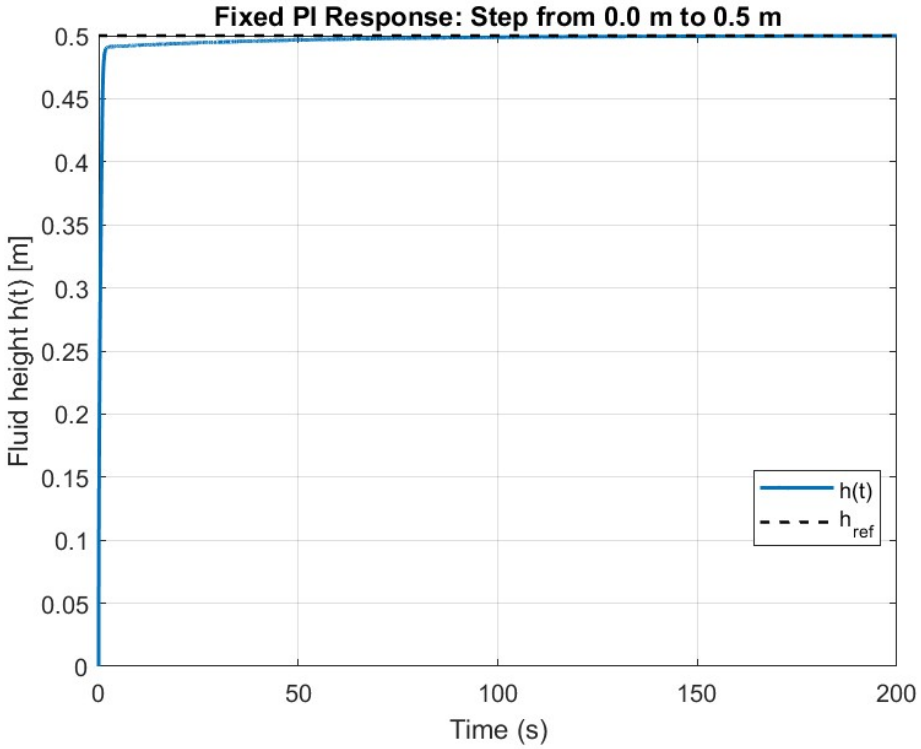


Figure 6: Fixed-gain PI response for the step from 0.0 m to 0.5 m.

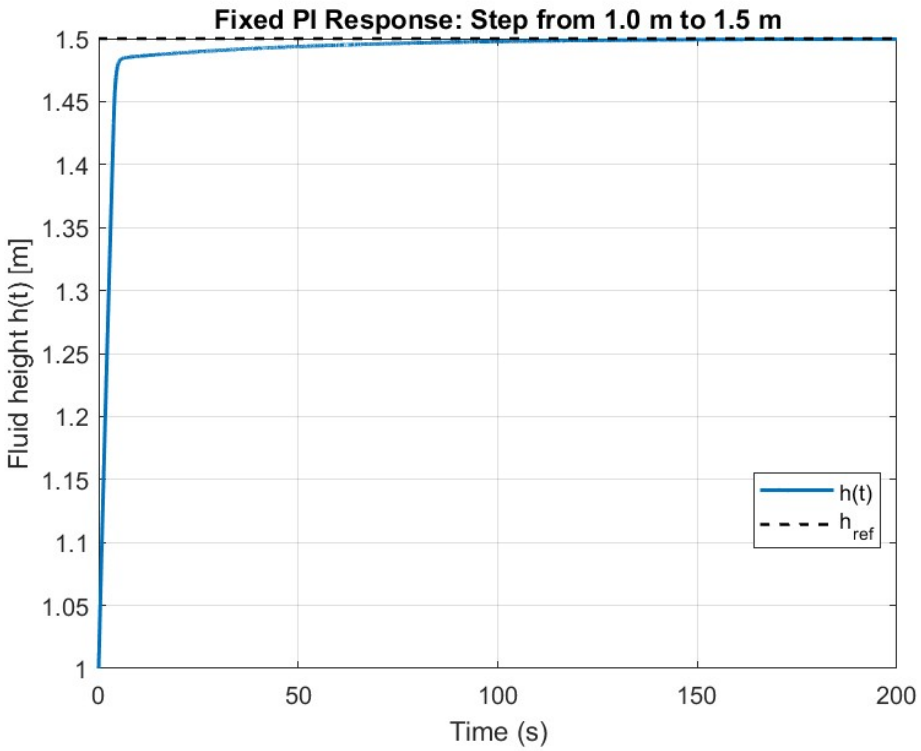


Figure 7: Fixed-gain PI response for the step from 1.0 m to 1.5 m using the same gains.

Table 4: Fixed-gain PI performance comparison.

Scenario	Final value	Steady-state error	Rise time	Settling time
$0.0 \rightarrow 0.5$ m	0.49983	0.00017271	0.96712 s	1.8994 s
$1.0 \rightarrow 1.5$ m	1.4997	0.00030933	3.5628 s	26.913 s

Although the same fixed PI gains achieved a fast and stable response for the low-level step, the response became much slower for the middle operating region. The rise time increased from 0.96712 s to 3.5628 s, and the settling time increased from 1.8994 s to 26.913 s. This degradation is caused by the nonlinear geometry of the spherical tank. Around $h = 1.5$ m, the cross-sectional area is close to its maximum. Since the height derivative is inversely proportional to $A(h)$, the same net inflow produces a smaller rate of change in the fluid height.

2.3 Fuzzy Gain-Scheduled PI Controller Design

To improve performance over different operating regions, a fuzzy gain-scheduled PI controller was implemented. The control law remains PI-based:

$$u(t) = K_p(t)e(t) + K_i(t)\int_0^t e(\tau)d\tau. \tag{14}$$

However, $K_p(t)$ and $K_i(t)$ are updated online by a Mamdani fuzzy inference system.

The FIS inputs are the normalized error E and the normalized derivative of error ΔE:

$$E = N_e e(t), \qquad \Delta E = N_{\Delta e} \dot{e}(t), \tag{15}$$

with saturation to the interval $[-1, 1]$. The FIS outputs are normalized gain coefficients K_p' and K_i' in the interval $[0, 1]$. The physical gains are then computed using linear mappings:

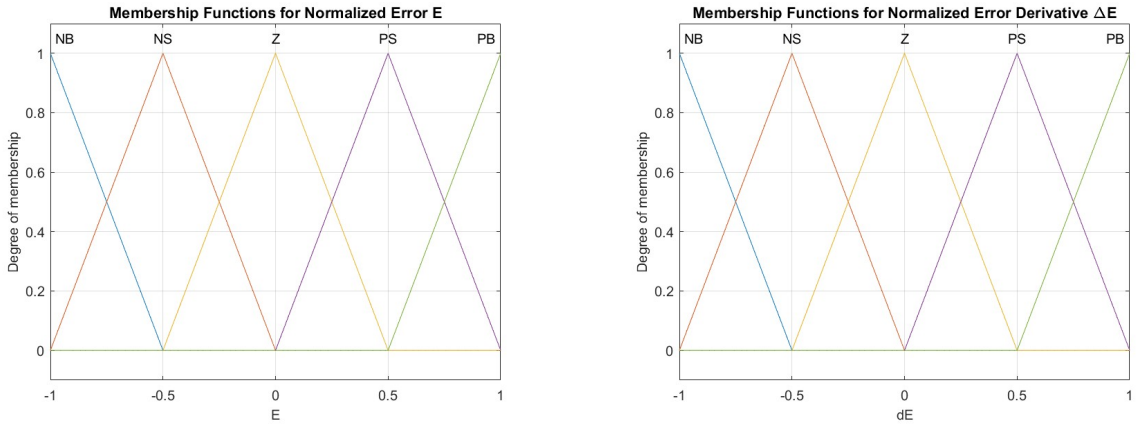
$$K_p(t) = K_{p,\min} + (K_{p,\max} - K_{p,\min}) K_p'(t), \tag{16}$$

$$K_i(t) = K_{i,\min} + (K_{i,\max} - K_{i,\min}) K_i'(t). \tag{17}$$

The gain bounds used in the simulation were

$$K_p \in [1, 20], \qquad K_i \in [0.01, 2]. \tag{18}$$

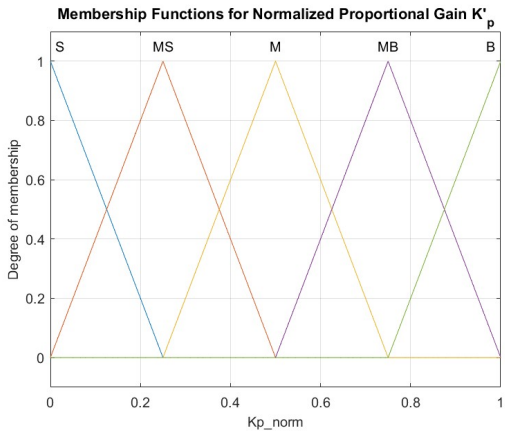
Each input was represented by five uniformly distributed triangular membership functions: NB, NS, Z, PS, and PB. Each output was represented by five triangular membership functions: S, MS, M, MB, and B. The FIS used min for the AND operation, max for aggregation, and centroid defuzzification.



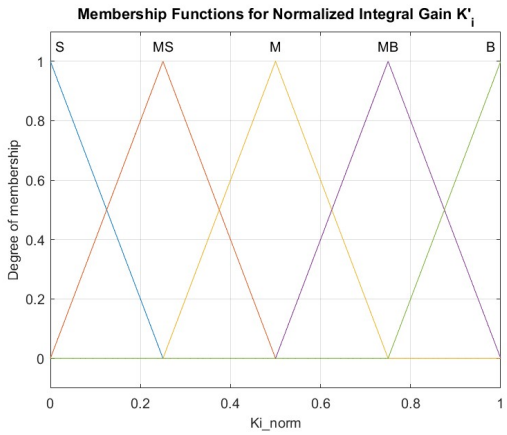
(a) Membership functions of E.

(b) Membership functions of ΔE.

Figure 8: Input membership functions of the FGS-PI fuzzy inference system.



(a) Membership functions of K'_p.



(b) Membership functions of K'_i.

Figure 9: Output membership functions of the FGS-PI fuzzy inference system.

Table 5: Rule base for K'_p. Rows correspond to ΔE and columns correspond to E.

$\Delta E \backslash E$	NB	NS	Z	PS	PB
NB	B	B	MB	M	S
NS	B	MB	M	S	M
Z	MB	M	S	M	MB
PS	M	S	M	MB	B
PB	S	M	MB	B	B

Table 6: Rule base for K'_i. Rows correspond to ΔE and columns correspond to E.

$\Delta E \backslash E$	NB	NS	Z	PS	PB
NB	S	S	S	M	B
NS	S	S	M	B	M
Z	S	M	B	M	S
PS	M	B	M	S	S
PB	B	M	S	S	S

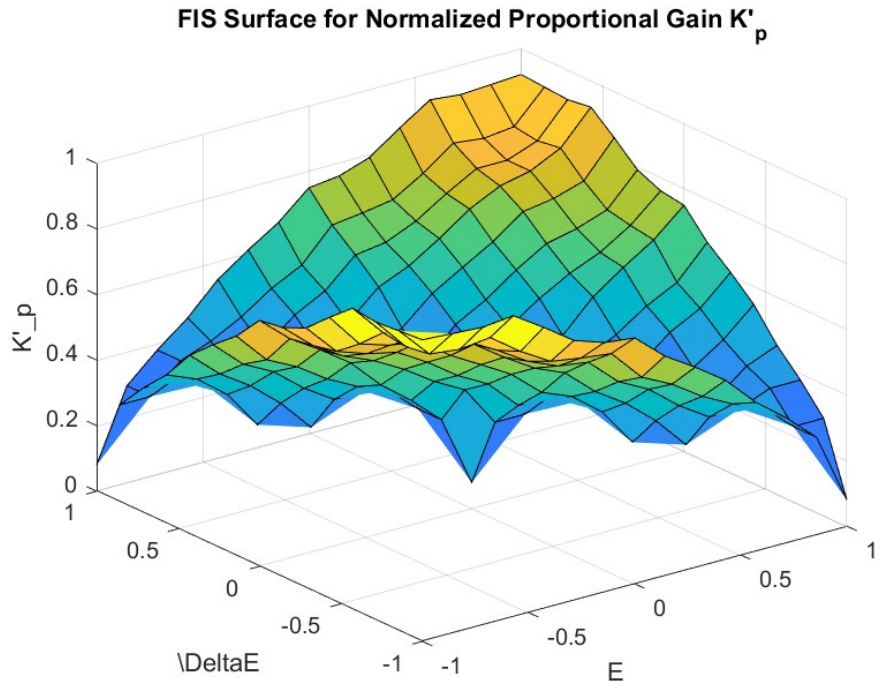


Figure 10: FIS surface for the normalized proportional gain K'_p.

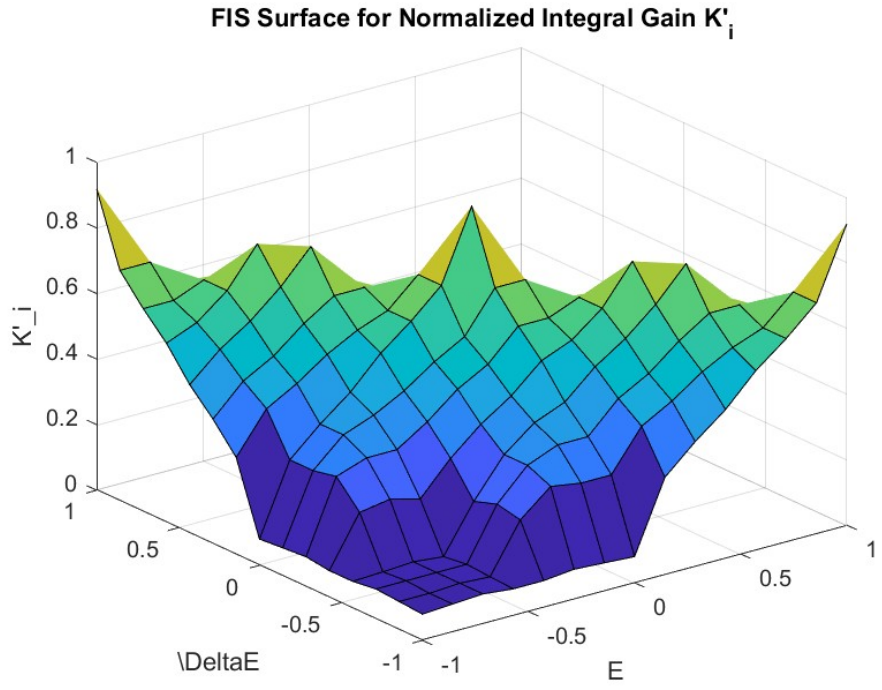


Figure 11: FIS surface for the normalized integral gain K'_i.

2.4 Fuzzy Control in Action

The fixed-gain PI controller was replaced by the fuzzy gain-scheduled PI controller. The reference signal was selected as a sequence of three setpoints, each lasting 200 s:

$$h_{\mathrm{ref}}(t) = \begin{cases} 0.5 \text{ m}, & 0 \le t < 200 \text{ s}, \\ 1.5 \text{ m}, & 200 \le t < 400 \text{ s}, \\ 2.5 \text{ m}, & 400 \le t \le 600 \text{ s}. \end{cases} \tag{19}$$

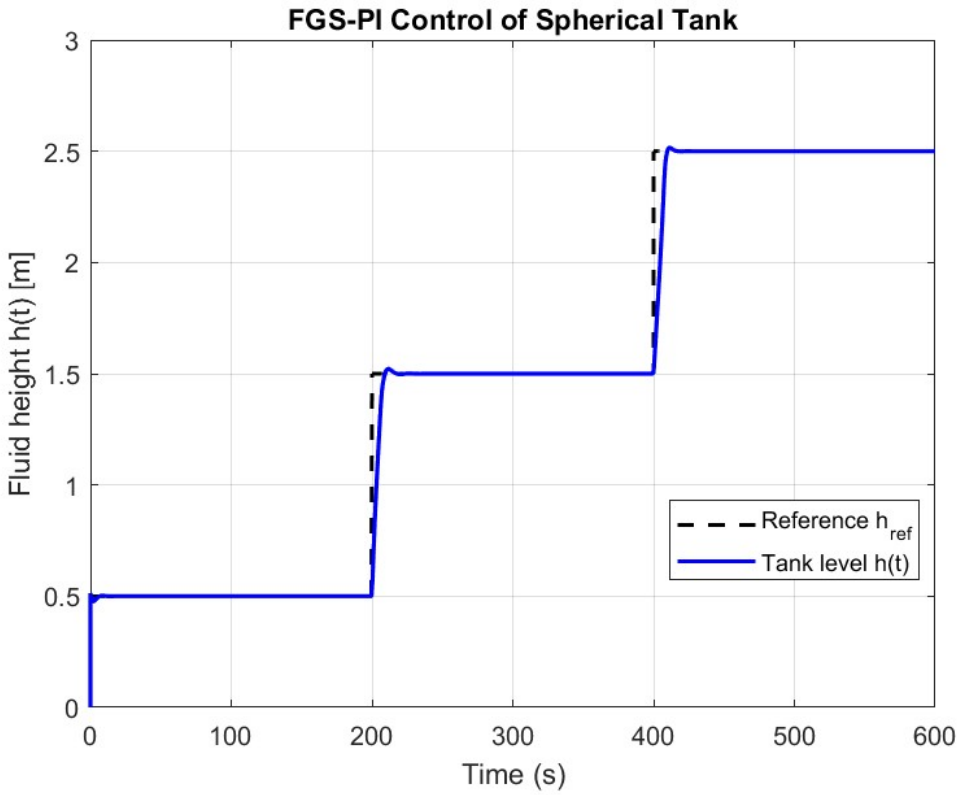


Figure 12: FGS-PI tracking response for the sequence of reference levels.

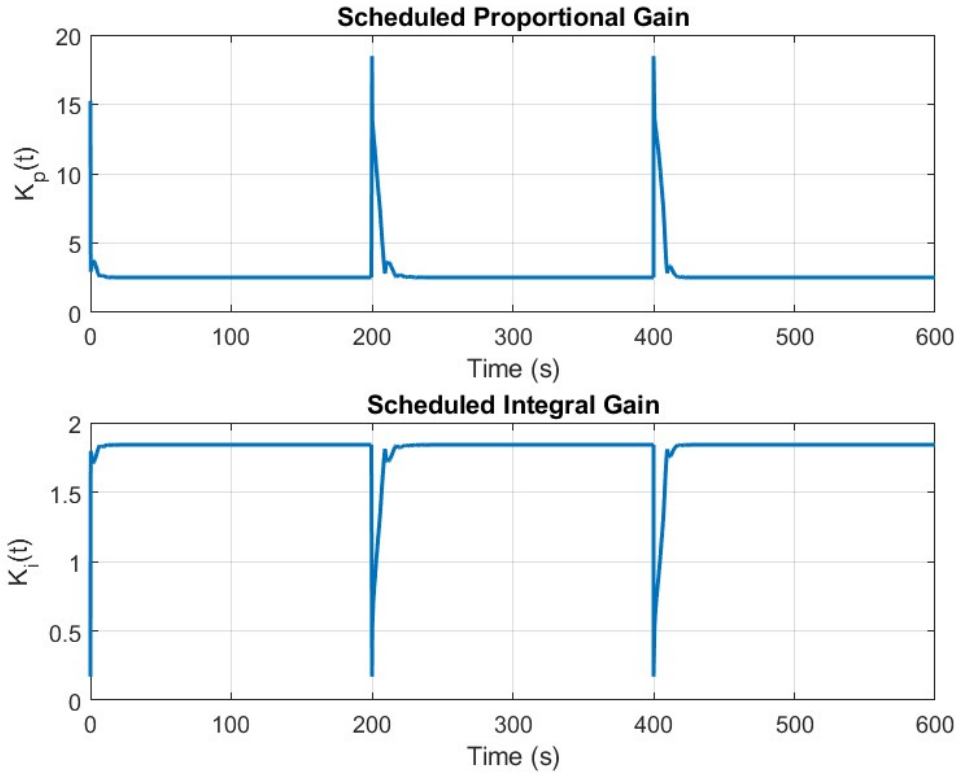


Figure 13: Scheduled proportional and integral gains, $K_p(t)$ and $K_i(t)$.

Table 7: FGS-PI tracking RMSE values for each reference interval.

Reference interval	RMSE
$h_{\text{ref}} = 0.5$ m	0.004508
$h_{\text{ref}} = 1.5$ m	0.102640
$h_{\text{ref}} = 2.5$ m	0.123800

The FGS-PI controller successfully tracks all three reference levels. The larger RMSE values in the second and third intervals are mainly caused by the transient periods immediately after the reference changes. Once the response approaches the new setpoint, the steady-state tracking error becomes very small.

The scheduled gain plots show that $K_p(t)$ increases sharply at the setpoint transition instants. This is expected because the tracking error becomes large immediately after each reference change. The increased proportional gain produces a faster corrective action. After the tank level approaches the reference, $K_p(t)$ decreases to reduce aggressiveness near steady state.

The integral gain $K_i(t)$ decreases immediately after abrupt reference changes and then increases again as the response approaches the desired level. This prevents excessive integral accumulation during large transient errors while still providing sufficient integral

action to eliminate steady-state error. Therefore, the fuzzy gain scheduling mechanism adapts the controller gains online according to the instantaneous error and error derivative, improving the performance over the nonlinear operating range of the tank.

3 Question 3: High-Level Fuzzy Speed Control for USV

3.1 Problem Description and USV Kinematics

The third problem considers waypoint navigation of an unmanned surface vessel (USV). The vessel is modeled by the non-holonomic kinematic equations

$$\dot{x} = v\cos\theta, \tag{20}$$

$$\dot{y} = v\sin\theta, \tag{21}$$

$$\dot{\theta} = \omega, \tag{22}$$

where (x, y) is the vessel position, θ is the heading angle, v is the forward speed, and ω is the angular velocity.

The desired heading angle toward the active waypoint is

$$\theta_d = \text{atan2}(y_{wp} - y, x_{wp} - x), \tag{23}$$

where (x_{wp}, y_{wp}) denotes the active waypoint. The heading error is wrapped to the interval $[-\pi, \pi]$:

$$\Delta\theta = \text{atan2}\left(\sin(\theta_d - \theta), \cos(\theta_d - \theta)\right). \tag{24}$$

The low-level steering controller is proportional:

$$\omega = K_h \Delta\theta, \tag{25}$$

with $K_h = 1.0$.

The fuzzy supervisory controller is designed only for the forward speed v.

3.2 Baseline Simulation and Orbiting Problem

The baseline controller uses a constant forward speed:

$$v = 2.0 \text{ m/s}. \tag{26}$$

To create a challenging scenario, the waypoint scale was reduced and the acceptance radius was made small:

$$wp_{\text{scale}} = 0.35, \qquad r_{\text{acc}} = 0.15 \text{ m}. \tag{27}$$

Under these conditions, the constant-speed controller fails to complete the waypoint circuit.

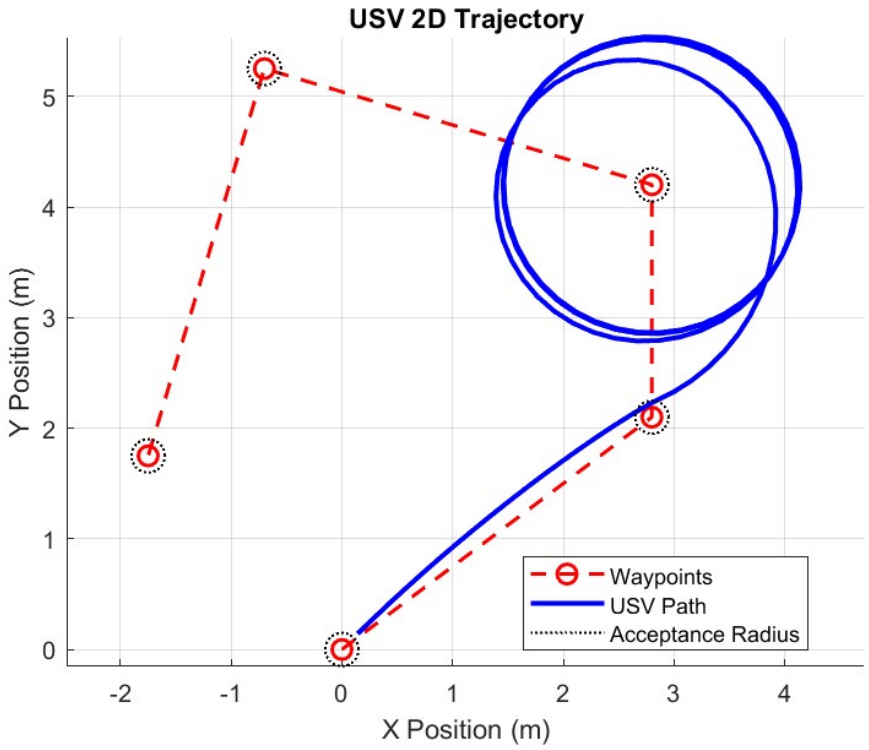


Figure 14: Baseline USV trajectory with constant speed. The USV enters an orbiting motion near the third waypoint.

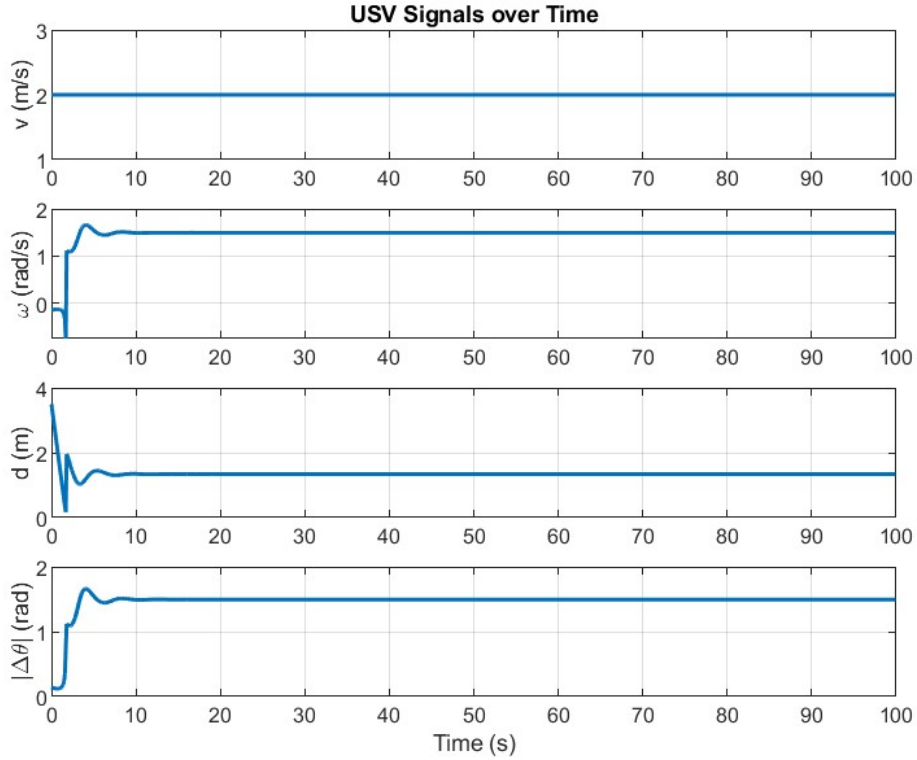


Figure 15: Baseline USV signals: constant speed, angular velocity, distance to waypoint, and absolute heading error.

Table 8: Baseline USV result.

Quantity	Value
Final waypoint index reached	3/5
Final distance to active waypoint	1.3381 m
Acceptance radius	0.1500 m
Simulation end time	99.90 s

The result shows that the USV successfully passes the second waypoint but fails to enter the acceptance region of the third waypoint. The final distance to the active waypoint is much larger than the acceptance radius. This confirms the orbiting problem.

The physical reason is the non-holonomic nature of the vessel. The USV cannot move laterally toward the waypoint; it can only move in the direction of its heading. When the forward speed is high and the heading error is large, the effective turning radius becomes large. Thus, the vessel overshoots the waypoint and keeps rotating around it rather than entering the small acceptance circle.

3.3 Fuzzy Rule Base Design

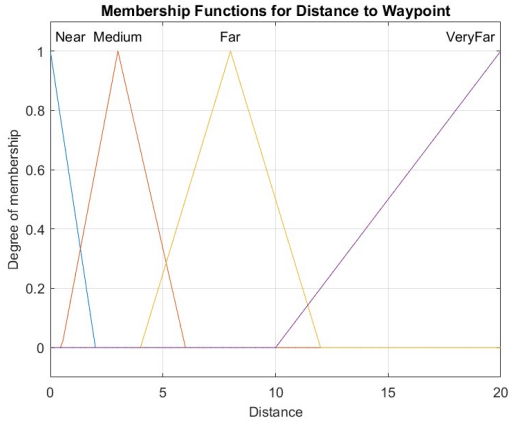
A Mamdani fuzzy inference system was designed to compute the forward speed v from two inputs:

- Distance to waypoint, $d \in [0, 20]$ m,
- Absolute heading error, $|\Delta\theta| \in [0, \pi]$ rad.

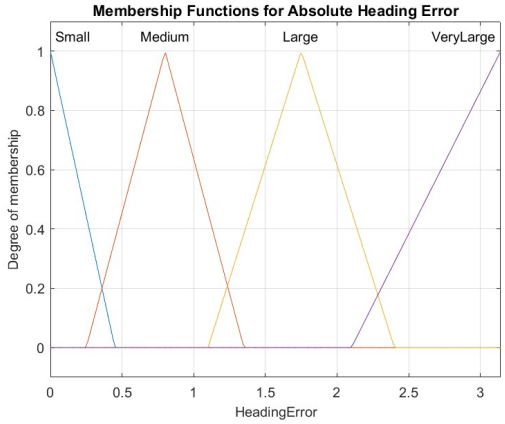
The output is the linear speed

$$v \in [0, 2] \text{ m/s}. \tag{28}$$

The design logic is based on intuitive navigation behavior. If the vessel is far from the waypoint and aligned with the target, it should move quickly. If it is close to the waypoint or has a large heading error, it should slow down to reduce the turning radius and avoid orbiting.



(a) Distance membership functions.



(b) Heading error membership functions.

Figure 16: Input membership functions of the USV speed controller.

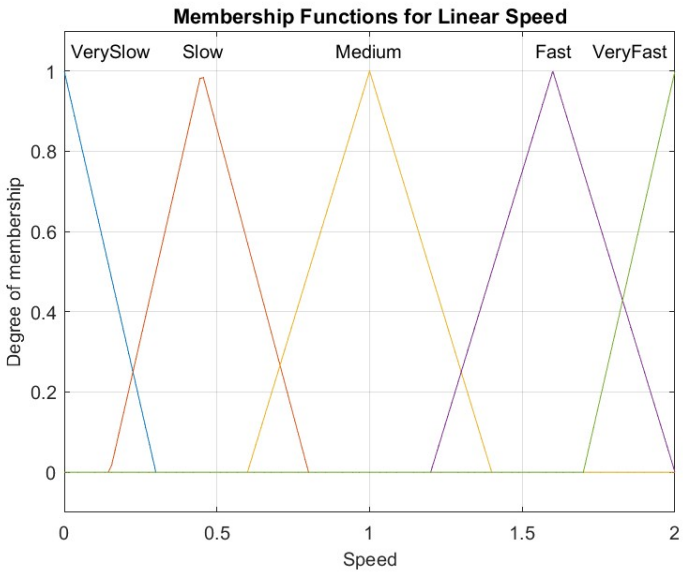


Figure 17: Output membership functions for linear speed.

Table 9: Fuzzy speed-control rule table.

Distance \ Heading Error	Small	Medium	Large	Very Large
Near	Slow	Slow	Very Slow	Very Slow
Medium	Fast	Medium	Slow	Very Slow
Far	Very Fast	Fast	Medium	Slow
Very Far	Very Fast	Very Fast	Fast	Medium

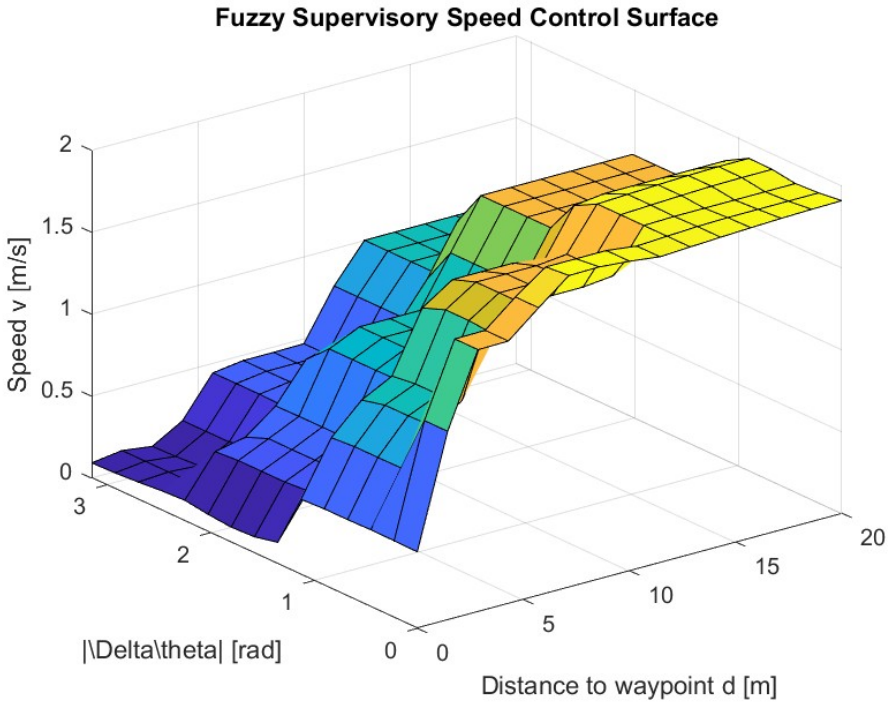


Figure 18: Fuzzy speed-control surface as a function of distance and absolute heading error.

3.4 Fuzzy Supervisory Control in Action

The fuzzy speed controller was then used under the same challenging waypoint configuration as the baseline case. The low-level heading controller remained unchanged:

$$\omega = K_h \Delta\theta. \tag{29}$$

Only the speed command was replaced by the fuzzy output:

$$v = F_{\mathrm{FIS}}(d, |\Delta\theta|). \tag{30}$$

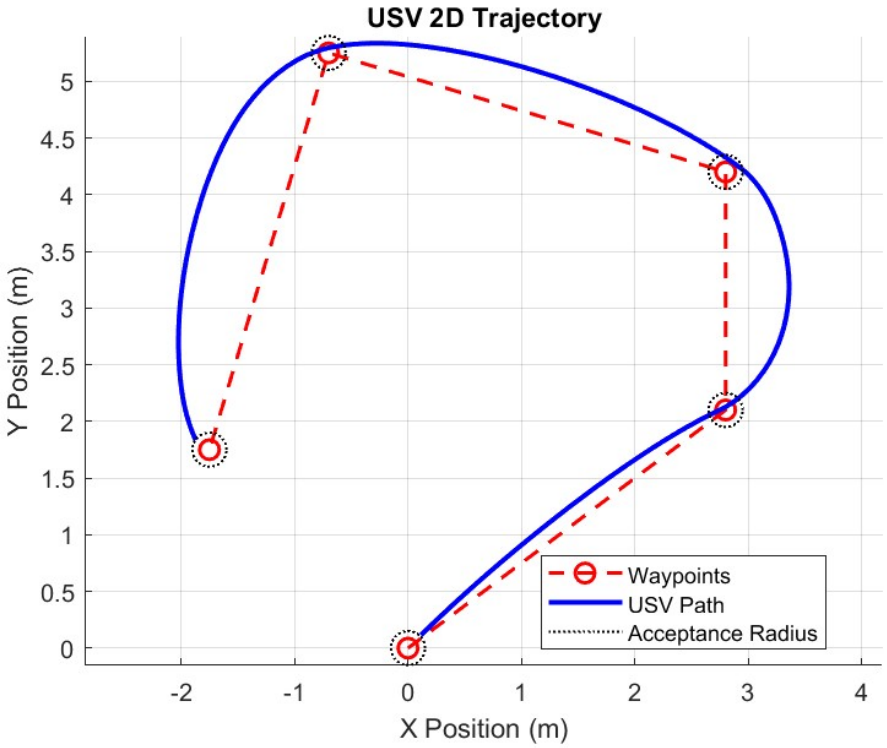


Figure 19: USV trajectory with the fuzzy supervisory speed controller. The waypoint circuit is completed successfully.

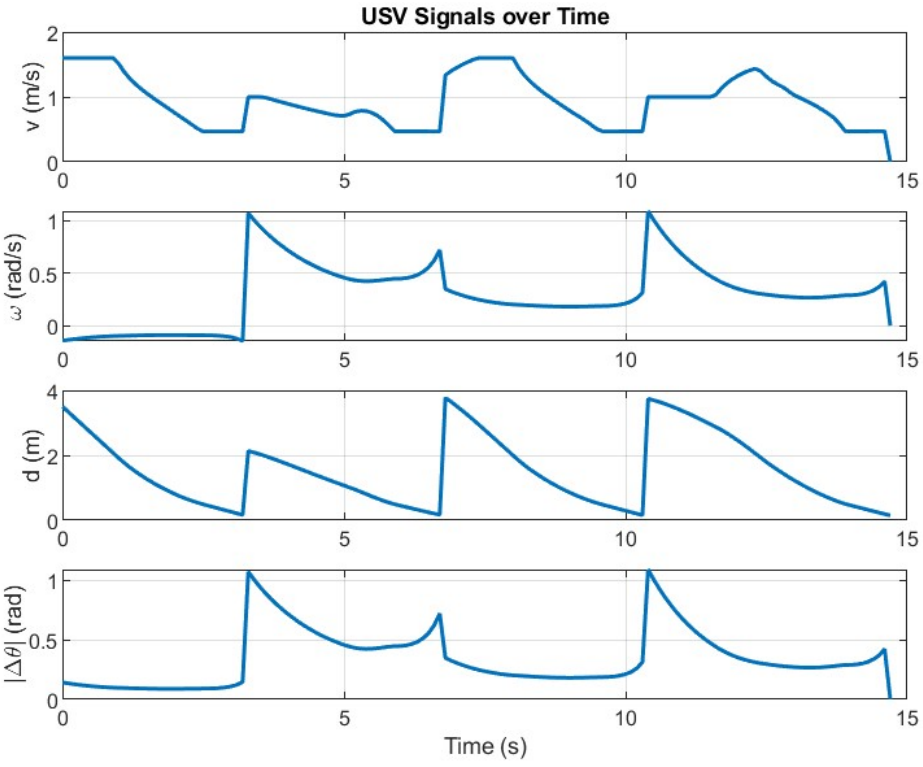


Figure 20: USV signals with fuzzy speed control: speed, angular velocity, distance to waypoint, and absolute heading error.

Table 10: Fuzzy-controlled USV result.

Quantity	Value
Final waypoint index reached	5/5
Final distance to active waypoint	0.1478 m
Acceptance radius	0.1500 m
Simulation end time	14.70 s

The fuzzy controller successfully completes the waypoint circuit. In contrast to the baseline simulation, the final waypoint index is 5/5, and the final distance to the active waypoint is 0.1478 m, which is slightly below the acceptance radius of 0.15 m.

The time-domain signals show that the speed is no longer constant. The speed decreases when the vessel is close to a waypoint or when the absolute heading error increases. At waypoint switching instants, $|\Delta\theta|$ increases because the target direction changes abruptly. The fuzzy supervisor responds by reducing v, allowing the heading controller to reorient the vessel before accelerating again. This behavior captures the desired human-like navigation strategy: slow down during sharp turns and speed up when aligned with the target.

4 Conclusion

In this assignment, three intelligent control problems were investigated. In Question 1, a NARX neural network was used to identify a nonlinear black-box discrete-time system. The correct inclusion of the current input in the input delay vector was essential, and the final model with delay order $d = 6$ achieved very low validation errors.

In Question 2, a nonlinear spherical tank was controlled using both fixed-gain PI and fuzzy gain-scheduled PI control. The fixed-gain PI controller showed clear performance degradation at the wider section of the tank due to the increased cross-sectional area. The FGS-PI controller adapted $K_p(t)$ and $K_i(t)$ online and improved tracking over the full operating range.

In Question 3, the constant-speed USV controller produced orbiting around a waypoint under a small acceptance radius. A fuzzy supervisory speed controller was designed using distance and heading error as inputs. The fuzzy controller dynamically reduced the speed near waypoints and during large heading-error conditions, successfully eliminating orbiting and completing the waypoint circuit.

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